

A twenty-edge necessary condition for a rank-six Ryser counterexample

Research continuation, 7 October 2026. This strengthens the repository's nineteen-edge necessary bound. It is a finite obstruction and does not solve Ryser's conjecture or claim mathematical priority.

1. Statement and prior input

Theorem. Every finite simple intersecting six-partite six-uniform hypergraph with at most nineteen edges has a vertex cover of size at most five. Thus any rank-six intersecting counterexample to Ryser's bound needs at least twenty distinct edges.

Each edge contains exactly one vertex from each of six specified parts. There is no upper bound on part sizes. Vertices outside the union of edges are ignored. We use the established lower half of $f(6) = 13$, with a complete elementary proof included in Appendix A:

Input F. Every intersecting six-partite six-uniform hypergraph with at most twelve edges has a cover of size at most four.

Here $f(6)$ is the minimum edge count under $\tau \geq 5$, not the minimum under $\tau = 6$. Input F was proved by Abu-Khazneh–Pokrovskiy [AP, Theorem 1.1, Section 2.1], and independently by Aharoni–Barát–Wanless [ABW]. AP's twelve-edge proof is a hand classification, with the earlier $f(6) \geq 12$ bound of Mansour–Song–Yuster as an input. Its Lemma 2.9 is written for $\tau = 5$; its same greedy bounds and low-degree pair count also exclude $\tau = 6$ at twelve edges. Appendix A instead reproduces the short ABW twelve-edge argument and supplies the at-most-eleven reduction explicitly. No linearity assumption, solver certificate or external unproved lemma is needed by the full note.

2. Degrees and the small-excess reduction

Suppose H is a counterexample, with $N \leq 19$ distinct edges. Any edge is a six-cover, so $\tau(H) = 6$. Every part has at least six active vertices. Every active vertex has degree at least two: if v belonged only to E , then $E \setminus \{v\}$ would meet every edge and be a five-cover.

If a vertex has degree $d \geq N - 12$, deleting its incident edges leaves at most twelve edges. Input F covers the remainder with four vertices, giving a five-cover of H . Consequently

$$\Delta(H) \leq N - 13. \tag{1}$$

Fixing an edge and counting other edges through its six vertices gives

$$N - 1 \leq 6(\Delta - 1) \leq 6(N - 14),$$

so $N \geq 17$.

Let

$$S = \sum_v \binom{d(v)}{2} = \sum_{\{E,F\}} |E \cap F|, \quad I = S - \binom{N}{2} \geq 0. \quad (2)$$

In each part its positive degrees sum to N , have at least six entries and lie between two and $N - 13$. For $N = 17$, this part energy is at most 18; for $N = 18$, at most 24. One direct bound is the chord inequality

$$\binom{d}{2} \leq 1 + \frac{M+1}{2}(d-2), \quad 2 \leq d \leq M,$$

which follows from $(d-2)(M-d) \geq 0$. For a part with $u \geq 6$ entries this gives energy at most $(M+1)N/2 - Mu$. With $(N, M) = (17, 4)$ the bound is 18.5, hence at most 18 by integrality; with $(N, M) = (18, 5)$ it is 24. Therefore

$$S \leq 108 < \binom{17}{2} \quad \text{or} \quad S \leq 144 < \binom{18}{2},$$

contradicting (2). Only $N = 19$ remains, with $\Delta \leq 6$.

For this case a part with u entries has $19 - 2u$ units of excess above two, each entry allowing at most four units. Convexity maximizes the sum of squared excesses by filling entries successively to capacity. Explicitly, transfer one unit from a positive smaller entry to a larger unsaturated entry: the squared sum increases by $2(a-b+1) > 0$. This proves the following complete small table:

active vertices in the part	maximum pair energy, degree at most 6
6	29
7	23
8	17
9	11

The first maximum is attained by $(6, 5, 2, 2, 2, 2)$. If the part has no degree-six vertex, its maximum is 26, attained by $(5, 5, 3, 2, 2, 2)$. Hence

$$S \leq 174, \quad I \leq 3. \quad (3)$$

If any part has no degree-six vertex, (2) forces exactly one such part: two would give $S \leq 168 < 171$. With one, $S \leq 171$, so $I = 0$. The other five parts have degree-six vertices. Their five incidence blocks have size six and pairwise intersections at most one, by linearity. Their union has size at least $5 \cdot 6 - \binom{5}{2} = 20 > 19$, a contradiction. Thus choose a degree-six vertex v_i in each part, $i = 1, \dots, 6$.

3. Four-vertex budgets force an impossible overlap pattern

Let B_i be the six original edges incident with v_i , and let $b_{ij} = |B_i \cap B_j|$. For any four-element $A \subset \{1, \dots, 6\}$, their union has size at most 16: if it covered at least seventeen original edges, the at most two remaining edges could be covered by one intersection vertex. That would give a five-cover. The elementary union lower bound therefore gives

$$16 \geq \left| \bigcup_{i \in A} B_i \right| \geq 24 - \sum_{\{i,j\} \subset A} b_{ij}, \quad \sum_{\{i,j\} \subset A} b_{ij} \geq 8. \quad (4)$$

Sum (4) over the fifteen four-subsets. Every pair occurs in six of them, so

$$\sum_{i < j} b_{ij} \geq 20, \quad \sum_{i < j} \binom{b_{ij}}{2} \geq 20 - 15 = 5. \quad (5)$$

The second inequality uses $\binom{b}{2} \geq b - 1$ for every nonnegative integer b , including zero.

Double counting pairs of vertices shared by pairs of original edges gives

$$Q := \sum_{\substack{\{v,w\} \\ v,w \text{ in different parts}}} \binom{|B_v \cap B_w|}{2} = \sum_{\{E,F\}} \binom{|E \cap F|}{2} \geq 5. \quad (6)$$

The lower bound follows from (5) by retaining only the selected vertices. Under $I \leq 3$, the only way to have $Q \geq 5$ is that exactly one original edge pair shares four vertices and all other edge pairs share exactly one. Indeed an edge-pair excess $j = |E \cap F| - 1$ contributes $j(j+1)/2$ to Q . Total excess at most three gives at most 4 unless it is the single term $j = 3$, which gives 6. This is the complete integer list, not a floating point estimate.

Let U be the four parts containing the four shared vertices of that exceptional pair. For any i, j , a value $b_{ij} \geq 2$ would give two original edges sharing v_i, v_j ; that edge pair must be the exceptional pair. Thus $b_{ij} \leq 2$, and a value two is possible only when $i, j \in U$. Choose four parts consisting of the two outside U and two inside U . Among their six pairs at most one can have codegree two; all others have codegree at most one. Their sum is at most seven, contradicting (4). This finishes the proof. $\square$

4. The next finite search case

An assumed twenty-edge counterexample satisfies $\Delta \leq 7$ by Input F, every active degree is at least two, and every part has at least six active vertices. The same convexity calculation gives the following maxima for one part:

active vertices	6	7	8	9	10
pair energy	35	29	22	14	10

Since $S \geq \binom{20}{2} = 190$, no part can have nine or more active vertices: even five other maximal parts give $5 \cdot 35 + 14 = 189 < 190$. Thus **eight vertices per part suffice for the complete twenty-edge case**. At most one part has eight vertices. If one does, at most one other part has seven; otherwise at most three parts have seven. These assertions follow by comparing the deficits from $6 \cdot 35 = 210$ with the budget of twenty.

Deleting any edge from a twenty-edge counterexample leaves nineteen edges, so the new theorem supplies a five-cover of the remainder. It must avoid the deleted edge and have exactly five vertices: otherwise adding one vertex of that edge gives a five-cover of all twenty edges. This justifies edge-critical search constraints at twenty edges. It is not a nonexistence theorem for that case.

5. Verification and scope

The proof is finite counting, including the prior Input F's full proof. `check_exact.py` enumerates the complete degree-partition domains independently, every excess partition up to three, all possible exceptional four-part supports, and the four-subset incidence count. It includes an incorrect-energy

negative control. The enumeration is a diagnostic companion to the complete inequalities above, not an unchecked solver status. Partial Lean exports verify the chord, integer codegree inequality and scalar budget transfers; they do not formalize hypergraph covers or the full appendix.

The earlier nineteen-edge package remains unchanged. This note does not exclude twenty-edge examples, establish the unrestricted rank-six case, or change the unrelated general Erdős–Lovász edge minimum. In particular it does not assert a general nonpartite $g(6) \geq 20$.

Appendix A. The prior twelve-edge four-cover bound

This appendix is attributed to [ABW, Theorem 2.7] for its twelve-edge classification. These are known ingredients, not claimed as new results. We give all the reductions needed here so the present twenty-edge proof has no external mathematical black box.

A.1. Seven-edge three-cover lemma. Every intersecting six-partite family with at most seven edges has a cover of size at most three. For at most six edges use pairing. With seven edges, a vertex of degree at least three, followed by pairing the remaining at most four edges, gives a three-cover. Otherwise all degrees are at most two. If no three-cover existed, every part would have at least four active vertices. In a part with total degree seven and at least four vertices of degrees one or two, at most three vertices have degree two. Thus the total pair energy would be at most $6 \cdot 3 = 18 < \binom{7}{2} = 21$, a contradiction. $\square$

A.2. At most eleven edges. Suppose a family has $N \leq 11$ edges and no four-cover. By A.1, $N \geq 8$. Selecting any vertex and applying A.1 to the remaining edges shows $\Delta \leq N - 8$. A fixed original edge then gives $N - 1 \leq 6(\Delta - 1) \leq 6(N - 9)$, hence $N \geq 11$. So $N = 11$, $\Delta \leq 3$, and every part has at least five active vertices. For degrees one through three, $\binom{d}{2} \leq 3(d - 1)/2$. Each part therefore has energy at most $3(11 - 5)/2 = 9$, and total energy at most $54 < \binom{11}{2} = 55$. This excludes the assumed family.

A.3. Twelve edges (ABW’s argument). Suppose $N = 12$ and no four-cover exists. A.1 shows $\Delta \leq 4$, because a degree at least five would leave at most seven edges. Let x_i count degree- i vertices and n be the active-vertex count. Every part has at least five active vertices, so $n \geq 30$.

There cannot be two degree-four vertices in one part: they cover eight distinct edges with two vertices and leave four edges to pair. Hence $x_4 \leq 6$. A part with no degree-four vertex has at most three degree-three vertices: four would cover all twelve edges. A part with a degree-four vertex has at most one degree-three vertex: two together with the degree-four vertex cover ten edges with three vertices, leaving a pair. Thus $x_3 \leq 18 - 2x_4$. The exact degree identity is

$$S = 72 - n + x_3 + 3x_4.$$

Consequently

$$66 \leq S \leq 42 + x_3 + 3x_4 \leq 60 + x_4 \leq 66.$$

All inequalities are equalities. In particular $I = 0$, and there is exactly one degree-four vertex in each part.

Any two of these six vertices have incident blocks that intersect: if disjoint, they cover eight edges, and pairing the remaining four gives a four-cover. By linearity they intersect in exactly one original edge. The fifteen pairs of high vertices have just twelve original edges to witness them, so some original edge contains at least three of the six vertices. Choose three on that edge. Their size-four incidence blocks meet pairwise exactly in that common edge, and their union has size $12 - 3 + 1 = 10$. Selecting

the three vertices and pairing the two remaining edges gives a four-cover, the final contradiction. This proves Input F. $\square$

References

[AP] A. Abu-Khazneh and A. Pokrovskiy, *Intersecting extremal constructions in Ryser’s Conjecture for r -partite hypergraphs*, <https://arxiv.org/abs/1409.4938v1>, Theorem 1.1 and Section 2.1. This source is cited as a preprint; no unverified journal reference is attached to it.

[ABW] R. Aharoni, J. Barát and I. M. Wanless, *Multipartite hypergraphs achieving equality in Ryser’s conjecture*, *Graphs and Combinatorics* **32** (2016), 1–15; <https://arxiv.org/abs/1409.4833v2>, DOI <https://doi.org/10.1007/s00373-015-1575-9>.

[Previous] The repository’s *Nineteen-edge necessary bound*, the rank-six research document `NINETEEN_EDGE_BOUND.md`, frozen at `db0b1c7`. Its proof does not use Input F. The stronger conclusion here uses that external input and a new four-vertex budget obstruction.